

Scientifically Defensible Explanation

Complete every numbered item. Correct this same PDF after Lesson Review.

Student name:

Evidence set

A sealed sample melts at 63 degrees C, freezes at 63 degrees C, and has a mass of 4.50 g before and after. A particle diagram shows the same particles arranged differently in solid and liquid states.

1-4. MODEL, LAW, THEORY, AND HYPOTHESIS ORGANIZER

1. MODEL - representation and limitation

2. LAW - conservation-of-mass pattern

3. THEORY - particle explanation for melting and freezing

4. HYPOTHESIS - testable prediction about reheating

Evidence Organizer and Recommendation

Use at least three exact pieces of evidence.

5. CHOOSE THE DEFENSIBLE EXPLANATION

Explanation A: Heating created a new substance because the sample changed from solid to liquid.

Explanation B: Heating caused a physical phase change; identity and mass remained the same.

6A. THREE EXACT PIECES OF EVIDENCE

Evidence 1

Evidence 2

Evidence 3

6B. ONE LIMITATION AND THE SUPPORTING SCIENTIFIC IDEA

State one limitation of the evidence or model and identify the law or theory that best supports your choice.

6C. SIX- TO EIGHT-SENTENCE RECOMMENDATION

Lesson Review Correction Record

Keep every original response visible.

AFTER LESSON REVIEW

Record each correction below. If no correction is needed, state that and explain what evidence confirms your work.

SUBMISSION CHECK

Save this corrected PDF and upload it in U1 L7 Notebook Evidence Submission.

Do not submit separate notebook pages for this Independent Application.